

CSS Font

Complete Guide: Styling Text with Typography

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1 Introduction

The CSS font properties control typography on web pages. Typography is fundamental to web design—it affects readability, user experience, visual hierarchy, and brand identity. Mastering font properties is essential for creating professional, engaging web interfaces.

1.1 Typography Importance

Good typography:

- Improves readability and comprehension
- Establishes visual hierarchy
- Reflects brand personality
- Enhances user experience
- Affects accessibility
- Influences emotional response

1.2 Font Properties Overview

CSS provides multiple font-related properties:

- `font-family` - Typeface selection
- `font-size` - Text size
- `font-weight` - Text boldness
- `font-style` - Italic or normal
- `font-variant` - Small caps and variants
- `line-height` - Line spacing
- `letter-spacing` - Character spacing
- `font` - Shorthand property

2 Font-Family Property

The `font-family` property specifies which typeface to use for text.

2.1 Font Family Values

```
/* Single font */
body {
  font-family: Arial;
}

/* Font stack (fallbacks) */
body {
  font-family: 'Helvetica Neue', Helvetica, Arial, sans-serif;
}

/* Generic families */
body {
  font-family: serif;
  font-family: sans-serif;
  font-family: monospace;
  font-family: cursive;
  font-family: fantasy;
}
```

2.2 Font Stacks

A font stack provides fallback options if the primary font isn't available:

```
body {
  font-family: 'Georgia', 'Times New Roman', serif;
}

/* Browser tries:
1. Georgia
2. Times New Roman (if Georgia not available)
3. Any serif font (generic fallback)
*/
```

2.3 Generic Font Families

Family	Description	Examples
serif	Has decorative lines	Georgia, Times New Roman
sans-serif	No decorative lines	Arial, Helvetica, Verdana
monospace	Fixed width	Courier, Monaco, Consolas
cursive	Handwriting style	Brush Script, Comic Sans
fantasy	Decorative	Impact, Papyrus

2.4 Web-Safe Fonts

These fonts are available on most systems:

```
/* Web-safe serif fonts */
font-family: 'Georgia', serif;
font-family: 'Times New Roman', serif;
font-family: 'Palatino', serif;

/* Web-safe sans-serif fonts */
font-family: 'Arial', sans-serif;
font-family: 'Helvetica', sans-serif;
font-family: 'Verdana', sans-serif;
font-family: 'Trebuchet MS', sans-serif;

/* Web-safe monospace fonts */
font-family: 'Courier New', monospace;
font-family: 'Courier', monospace;
font-family: 'Monaco', monospace;
```

2.5 Google Fonts

Import custom fonts from Google Fonts service:

```
/* In HTML head */
<link
  href="https://fonts.googleapis.com/css2?family=Roboto:wght@400;700&display=swap"
  rel="stylesheet">

/* In CSS */
body {
  font-family: 'Roboto', sans-serif;
}

h1 {
  font-family: 'Roboto', sans-serif;
  font-weight: 700;
}
```

3 Font-Size Property

The `font-size` property controls text size.

3.1 Size Units

Unit	Type	Use
px	Absolute	Fixed size, precise control
em	Relative	Relative to parent font size
rem	Relative	Relative to root font size
%	Relative	Relative to parent font size
pt	Print	Printer points, avoid web

3.2 Font-Size Examples

```
/* Pixel sizing */
body {
  font-size: 16px; /* Base size */
}

h1 {
  font-size: 32px; /* Double base */
}

/* EM sizing (relative to parent) */
body {
  font-size: 16px;
}

p {
  font-size: 1em; /* 16px */
}

p em {
  font-size: 1.5em; /* 24px (1.5 × 16px) */
}

/* REM sizing (relative to root) */
html {
  font-size: 16px; /* Base root size */
}

body {
  font-size: 1rem; /* 16px */
}

h1 {
  font-size: 2rem; /* 32px */
}

h2 {
  font-size: 1.5rem; /* 24px */
}

small {
  font-size: 0.875rem; /* 14px */
}
```

3.3 Responsive Font Sizing

```
/* Fluid typography */
body {
  font-size: clamp(14px, 2vw, 18px);
  /* Min: 14px, Preferred: 2% viewport width, Max: 18px */
}

/* Breakpoint-based */
```

```
body {  
  font-size: 16px;  
}  
  
@media (max-width: 768px) {  
  body {  
    font-size: 14px;  
  }  
}  
  
@media (min-width: 1200px) {  
  body {  
    font-size: 18px;  
  }  
}
```

4 Font-Weight Property

The `font-weight` property controls text boldness.

4.1 Font-Weight Values

Value	Number	Description
<code>normal</code>	400	Regular font weight
<code>bold</code>	700	Bold font weight
<code>100-900</code>	Numeric	Specific weight levels
<code>lighter</code>	Relative	Lighter than parent
<code>bolder</code>	Relative	Bolder than parent

4.2 Font-Weight Examples

```
/* Keyword values */
p {
  font-weight: normal; /* 400 */
}

strong {
  font-weight: bold; /* 700 */
}

/* Numeric values */
.light {
  font-weight: 300; /* Thin/Light */
}

.regular {
  font-weight: 400; /* Normal */
}

.semi-bold {
  font-weight: 600; /* Semi-bold */
}

.bold {
  font-weight: 700; /* Bold */
}

.extra-bold {
  font-weight: 800; /* Extra bold */
}

/* Relative values */
.lighter-text {
  font-weight: lighter; /* Lighter than parent */
}

.bolder-text {
  font-weight: bolder; /* Bolder than parent */
}
```



```
}
```

4.3 Variable Fonts

Modern variable fonts support any weight between defined ranges:

```
@font-face {
  font-family: 'Roboto Flex';
  src: url('RobotoFlex.woff2') format('woff2-variations');
  font-weight: 100 900; /* Full range */
}

body {
  font-family: 'Roboto Flex', sans-serif;
  font-weight: 400;
}

.thin {
  font-weight: 100; /* Available */
}

.custom {
  font-weight: 450; /* Any value available */
}
```

5 Font-Style Property

The `font-style` property controls italic styling.

5.1 Font-Style Values

Value	Description
<code>normal</code>	Upright text (default)
<code>italic</code>	Slanted italic text
<code>oblique</code>	Simulated slanted text

5.2 Font-Style Examples

```
/* Normal text */
p {
  font-style: normal;
}

/* True italic */
em {
  font-style: italic;
}

/* Oblique (simulated italic) */
.oblique-text {
  font-style: oblique;
}
```

```

}

/* Oblique with angle */
.oblique-15 {
  font-style: oblique 15deg; /* Modern browsers */
}

/* Removing italic */
em {
  font-style: italic;
}

em.normal {
  font-style: normal; /* Override italic */
}

```

6 Line-Height Property

The `line-height` property controls spacing between lines of text.

6.1 Line-Height Values

Type	Examples	Use
Number (unitless)	1, 1.5, 2	Relative to font size
Length	1.5em, 20px, 2rem	Specific measurements
Percentage	100%, 150%	Relative to font size
normal	Default	Browser default (1.2)

6.2 Line-Height Examples

```

/* Unitless (recommended) */
body {
  font-size: 16px;
  line-height: 1.5; /* 24px line height */
}

/* EM units */
p {
  font-size: 14px;
  line-height: 1.75em; /* 24.5px */
}

/* Pixel units */
h1 {
  font-size: 32px;
  line-height: 40px;
}

/* Percentage */
.dense {
  font-size: 16px;

```

```
    line-height: 120%; /* 19.2px */
}

.loose {
    font-size: 16px;
    line-height: 200%; /* 32px */
}

/* Normal keyword */
p {
    line-height: normal; /* Browser default */
}
```

6.3 Best Practices for Line-Height

```
/* Good for readability */
body {
    font-size: 16px;
    line-height: 1.6; /* 25.6px */
}

/* For headings */
h1, h2, h3 {
    line-height: 1.2; /* Tighter for headings */
}

/* For code blocks */
code {
    line-height: 1.4; /* Slightly loose */
}

/* For body text on mobile */
@media (max-width: 768px) {
    body {
        line-height: 1.7; /* Slightly looser */
    }
}
```

7 Font Shorthand

The `font` shorthand property combines multiple font properties.

7.1 Font Shorthand Syntax

```
font: [ font-style ] [ font-variant ] [ font-weight ] [ font-size / line-height ]  
      font-family;
```

7.2 Font Shorthand Examples

```
/* Minimal */  
p {  
  font: 16px Arial;  
}  
  
/* With weight and style */  
strong {  
  font: bold 18px Georgia;  
}  
  
/* With line-height */  
body {  
  font: 16px/1.6 'Helvetica Neue', Arial, sans-serif;  
}  
  
/* Full specification */  
h1 {  
  font: italic bold 32px/1.2 'Times New Roman', serif;  
}  
  
/* With variant */  
.small-caps {  
  font: small-caps 14px Georgia;  
}
```

7.3 Why Use Font Shorthand

- Concise and readable
- Resets all font properties
- Ensures consistency
- Reduces CSS file size

8 Letter-Spacing and Word-Spacing

Fine-tune character and word spacing.

8.1 Letter-Spacing

```
/* Normal spacing */
p {
  letter-spacing: normal;
}

/* Increase spacing */
.spaced {
  letter-spacing: 2px;
}

.loose-text {
  letter-spacing: 0.1em;
}

/* Decrease spacing */
.tight {
  letter-spacing: -1px;
}

/* Headings */
h1 {
  letter-spacing: 3px; /* Elegant spacing */
}
```

8.2 Word-Spacing

```
/* Normal spacing */
p {
  word-spacing: normal;
}

/* Increase spacing */
.spaced-words {
  word-spacing: 5px;
}

.loose-words {
  word-spacing: 0.2em;
}

/* Decrease spacing */
.tight-words {
  word-spacing: -2px;
}
```

9 Real-World Examples

9.1 Example 1: Professional Blog

```
html {
  font-size: 16px;
}

body {
  font-family: 'Georgia', 'Times New Roman', serif;
  font-size: 1rem;
  line-height: 1.6;
  color: #333;
}

h1 {
  font-family: 'Helvetica Neue', Arial, sans-serif;
  font-size: 2.5rem;
  font-weight: 700;
  line-height: 1.2;
  margin-bottom: 0.5rem;
}

h2 {
  font-size: 1.75rem;
  font-weight: 600;
  line-height: 1.3;
}

p {
  margin-bottom: 1rem;
}

blockquote {
  font-style: italic;
  font-size: 1.1rem;
  border-left: 4px solid #ddd;
  padding-left: 1rem;
  margin: 2rem 0;
}

code {
  font-family: 'Courier New', monospace;
  font-size: 0.9rem;
  background-color: #f4f4f4;
  padding: 2px 4px;
}
```

9.2 Example 2: Modern SaaS Landing Page

```
html {
  font-size: 16px;
}

body {
  font-family: -apple-system, BlinkMacSystemFont,
```

```
        'Segoe UI', Roboto, sans-serif;
font-size: 1rem;
line-height: 1.5;
color: #1f2937;
}

h1 {
  font-size: clamp(2rem, 5vw, 3.5rem);
  font-weight: 800;
  line-height: 1.1;
  letter-spacing: -0.5px;
}

.tagline {
  font-size: 1.25rem;
  font-weight: 300;
  letter-spacing: 0.5px;
  margin-top: 0.5rem;
}

button {
  font-family: inherit;
  font-weight: 600;
  font-size: 1rem;
}

.small-text {
  font-size: 0.875rem;
  color: #6b7280;
}
```

9.3 Example 3: E-commerce Product Page

```
body {
  font-family: -apple-system, BlinkMacSystemFont, sans-serif;
  font-size: 16px;
  line-height: 1.5;
}

.product-title {
  font-size: 2rem;
  font-weight: 700;
  line-height: 1.2;
}

.price {
  font-size: 1.5rem;
  font-weight: 600;
  color: #e11d48;
}

.description {
  font-size: 1rem;
```

```

    line-height: 1.6;
    color: #4b5563;
}

.rating {
    font-size: 0.9rem;
    font-weight: 500;
}

.badge {
    font-size: 0.75rem;
    font-weight: 700;
    text-transform: uppercase;
    letter-spacing: 1px;
}

```

10 Web Fonts and @font-face

Use custom fonts with @font-face.

10.1 @font-face Syntax

```

@font-face {
    font-family: 'CustomFont';
    src: url('font.woff2') format('woff2'),
         url('font.woff') format('woff'),
         url('font.ttf') format('truetype');
}

body {
    font-family: 'CustomFont', sans-serif;
}

```

10.2 Font File Formats

Format	Extension	Browser Support
WOFF2	.woff2	Modern browsers (95%+)
WOFF	.woff	All modern browsers
TTF/OTF	.ttf/.otf	Most browsers
EOT	.eot	IE 9+

10.3 Google Fonts Implementation

```

<!-- Link in HTML head -->
<link
    href="https://fonts.googleapis.com/css2?family=Inter:wght@400;600;700&display=swap"
    rel="stylesheet">

<!-- Use in CSS -->
body {

```



```
font-family: 'Inter', sans-serif;
}

.bold {
  font-weight: 700;
}
```

11 Font Performance

Optimize font loading and rendering.

11.1 Font Display Property

```
@font-face {
  font-family: 'CustomFont';
  src: url('font.woff2') format('woff2');
  font-display: swap; /* Show fallback while loading */
}

/* Options:
  - auto: Default browser behavior
  - block: Hide text until font loads
  - swap: Show fallback, replace when ready
  - fallback: Brief block, then swap
  - optional: Don't block, may not swap
*/
```

11.2 Variable Fonts

Variable fonts reduce file size:

```
@font-face {
  font-family: 'Roboto Flex';
  src: url('RobotoFlex.woff2') format('woff2-variations');
  font-weight: 100 900;
  font-style: normal italic;
}

/* One file handles all weights and styles */
body {
  font-family: 'Roboto Flex', sans-serif;
}

.light {
  font-weight: 300;
}

.bold {
  font-weight: 700;
}
```

```
.italic {  
  font-style: italic;  
}
```

12 Font Accessibility

Ensure fonts are accessible.

12.1 Readability Best Practices

```
body {  
  font-size: 16px; /* Minimum for body text */  
  line-height: 1.5; /* Good spacing */  
  color: #333; /* Good contrast */  
  background-color: #fff;  
}  
  
/* Accessible link styling */  
a {  
  color: #0066cc;  
  text-decoration: underline;  
  font-weight: 600;  
}  
  
/* Clear hierarchy */  
h1 {  
  font-size: 2rem;  
  font-weight: 700;  
}  
  
h2 {  
  font-size: 1.5rem;  
  font-weight: 600;  
}  
  
/* Avoid all-caps */  
.all-caps {  
  text-transform: uppercase;  
  letter-spacing: 1px; /* Add spacing for readability */  
}
```

13 Browser Support

13.1 Font Properties Support

Property	Chrome	Firefox	Safari	Edge
font-family	All	All	All	All
font-size	All	All	All	All
font-weight	All	All	All	All
font-style	All	All	All	All
line-height	All	All	All	All
@font-face	All	3.5+	3.1+	All
Variable fonts	62+	62+	11.1+	79+

14 Common Mistakes

14.1 Mistake 1: Ignoring Line-Height

Bad:

```
p {
  font-size: 16px;
  /* Line-height defaults to normal */
  /* Text appears cramped */
}
```

Good:

```
p {
  font-size: 16px;
  line-height: 1.6; /* Proper spacing */
}
```

14.2 Mistake 2: Too Many Font Families

Bad:

```
h1 { font-family: 'Font1', sans-serif; }
p { font-family: 'Font2', sans-serif; }
button { font-family: 'Font3', sans-serif; }
/* Performance suffers */
```

Good:

```
body {
  font-family: -apple-system, BlinkMacSystemFont, sans-serif;
}

h1 {
  font-family: inherit; /* Use body font */
  font-weight: 700;
}
```

14.3 Mistake 3: Poor Font Sizes

Bad:

```
p {  
  font-size: 12px;  /* Too small */  
}  
  
h1 {  
  font-size: 5rem;  /* Too large */  
}
```

Good:

```
body {  
  font-size: 16px;  
}  
  
p {  
  font-size: 1rem;  /* 16px */  
}  
  
h1 {  
  font-size: 2.5rem; /* 40px */  
}
```

15 Summary: Key Takeaways

15.1 Font Properties Quick Reference

Property	Purpose	Common Values
font-family	Typeface	Arial, Georgia, serif
font-size	Text size	16px, 1rem, 1.5em
font-weight	Boldness	400, 700, bold
font-style	Italic	normal, italic
line-height	Line spacing	1.6, 1.5, 24px
letter-spacing	Char spacing	2px, 0.1em

15.2 Best Practices

1. Use system fonts for performance
2. Set base font-size on html element
3. Use relative units (em, rem) for scalability
4. Maintain line-height between 1.4-1.7
5. Ensure sufficient color contrast
6. Choose 1-2 font families maximum
7. Test fonts across browsers and devices

8. Use variable fonts for flexibility
9. Optimize font loading
10. Maintain accessibility standards

15.3 Quick Reference

```
/* Good typography foundation */
html {
  font-size: 16px;
}

body {
  font-family: -apple-system, BlinkMacSystemFont, sans-serif;
  font-size: 1rem;
  line-height: 1.6;
  color: #333;
}

h1, h2, h3 {
  font-weight: 700;
  line-height: 1.2;
}

h1 {
  font-size: 2.5rem;
}

h2 {
  font-size: 2rem;
}

h3 {
  font-size: 1.5rem;
}

@media (max-width: 768px) {
  h1 {
    font-size: 2rem;
  }

  h2 {
    font-size: 1.5rem;
  }
}
```

16 Conclusion

CSS font properties are fundamental to web design. Mastering typography transforms ordinary web pages into professional, engaging experiences.

Key principles to remember:

1. Font-family: Choose readable, performant fonts
2. Font-size: Use relative units for scalability
3. Font-weight: Establish visual hierarchy
4. Line-height: Prioritize readability
5. Performance: Minimize font files and requests
6. Accessibility: Ensure sufficient contrast
7. Consistency: Maintain typographic hierarchy
8. Responsiveness: Adapt fonts to screen sizes
9. Testing: Validate across browsers and devices
10. User experience: Let typography guide the user

With strong typography skills, you'll create web experiences that are both beautiful and functional, improving user engagement and satisfaction.

Happy coding!